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When Dougall Tails Terminate

A rationality classification, Stieltjes cancellation, and an all-depth harmonic–polylogarithmic sum formula

Research continuation prepared for Vladimir Reshetnikov

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Abstract

We classify all rational and all finitely terminating centered Gamma quotients at the integral resonances of the positive-parameter Rogers–Dougall family. Rationality occurs precisely when one of three symmetric parameters is integral. Finite termination occurs precisely on the evident short-integer polynomial loci or on one exceptional symmetric Laurent-polynomial locus. A rational difference-equation argument shows why termination of the formal asymptotic series is an exact identity, not merely an asymptotic assertion. In contrast, analytic curves of nonrational summands can have identically zero summed remainder.

A two-parameter resolvent specialization then separates the Stieltjes boundary term from the convergent higher moments. Subtracting two resolvents cancels every Stieltjes coefficient and yields a Gamma expression for all fixed-order polygamma moments. At the half-center, a normalized Gamma quotient evaluates an infinite family of weighted composition sums in ordinary generalized harmonic numbers. The resulting all-depth level-two polylogarithmic sum formula has values in the algebra of ordinary zeta values and $\log 2$. Explicit cubic and quartic sums, a dilogarithmic integral, Euler primitives, and zero-safe derivative formulas are given with proofs and reproducible diagnostics.

The classical summation and the repository's prior all-resonance continuation are credited separately. Global literature priority is not established, and no proof-assistant formalization is claimed.

Repository snapshot: `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`
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1 Scope, antecedents, and the precise additions

The ProveIt manuscript organizes polylogarithms, harmonic sums, Gamma products, and Hurwitz-zeta derivatives by exact transformations rather than by numerical recognition alone. Its canonical mixed-Dougall section proves a first-resonance identity, fixed mixed derivatives, and a weighted trigamma square [6]. A preserved continuation already proves the centered all-resonance subtraction, its collapsed digamma value, and all spectral jets [7]. Those results are inputs and antecedents here, not newly solved conjectures.

The present work begins with a different question: *when is the subtracted tail actually zero term by term?* Several sufficient polynomial loci appear in the preserved report. We prove a complete classification within its positive-parameter domain. We then show that a vanishing *sum* of the tail is a substantially weaker condition: there are analytic families of nonrational Gamma quotients whose remainders sum to zero. This gives a concrete distinction between an exact finite summand identity and an exact infinite cancellation.

The second addition is a resolvent construction. A symmetric perturbation of the Gamma parameters produces the denominator $x_n(x_n^2 - v^2)$. Its Gamma evaluation is an ordinary convergent identity. In a companion boundary identity, every generalized Stieltjes constant can be displayed explicitly. Taking a resolvent difference removes all those constants at once. After half-center normalization, this yields a compact generating theorem for nonlinear harmonic sums, together with exact signed sums of multiple polylogarithms of arbitrary depth.

The Rogers–Dougall summation is classical [1], as are Gamma recurrences, Stirling expansions, and Hurwitz-zeta identities [2, 3]. Parameter differentiation of hypergeometric sums is an established source of harmonic and zeta identities. The Euler-sum literature includes extensive linear and quadratic evaluations and colored multiple-zeta representations; Xu and Wang provide one relevant modern reference [4]. Consequently, the statements below are proved mathematical results, but "proved here" does not mean "first in the literature." The rationality/termination classification, the nonterminating zero-remainder curves, and the specific unified resolvent/composition organization are the principal additions relative to the inspected repository text.

Audit limits. The source inspection covered the pinned canonical text and the specific preserved continuation listed in the source-audit file. The five ZIP archives present in `docs/incoming` were inventoried by name and metadata but could not be unpacked through the available binary-access route. Their contents were not checked for overlap. The external antecedent check used DLMF and the Xu–Wang publisher page, not an exhaustive literature review. The Gaussian S_6 conjecture is not resolved by this package. The manuscript's rejected frozen S_8 vector is not revived. These limitations do not weaken the proofs below, but they do limit priority and coverage claims.

2 Centered Dougall subtraction: the established starting point

Throughout the classification sections, fix

$$\kappa, b, c > 0, \qquad d_0 = 1 + 2\kappa - b - c > 0, \qquad x_n = n + \kappa. \tag{2.1}$$

All denominator Gamma factors mean entire reciprocal-Gamma functions. In particular, a zero reciprocal-Gamma value must not be replaced by a zero germ before differentiation. Define

$$W_n(u) = x_n \frac{\Gamma(n+2\kappa)\Gamma(n+b)\Gamma(n+c)\Gamma(n+d_0-u)}{\Gamma(n+1)\Gamma(n+1+2\kappa-b)\Gamma(n+1+2\kappa-c)\Gamma(n+b+c+u)}, \tag{2.2}$$

$$P(u) = \frac{\Gamma(b)\Gamma(c)\Gamma(d_0-u)}{\Gamma(d_0)\Gamma(b+u)\Gamma(c+u)}. \tag{2.3}$$

The classical Rogers–Dougall identity in this normalization is

$$\sum_{n\geq 0} W_n(u) = \frac{1}{2}\Gamma(u)P(u), \qquad 0 < \operatorname{Re} u < d_0. \tag{2.4}$$

The hypergeometric excess is $2u$. The factor $1/2$ is essential: the usual very-well-poised factor is $(n+\kappa)/\kappa$, whereas (2.2) uses x_n and unnormalized Gamma factors.

Put

$$\alpha(u) = (\kappa, b-\kappa, c-\kappa, 1+\kappa-b-c-u). \tag{2.5}$$

The denominator shifts relative to x_n are $1-\alpha_i(u)$. Let $B_j(x)$ denote Bernoulli polynomials and define

$$\ell_j(u) = -\frac{\sum_{i=1}^{4} B_{2j+1}(\alpha_i(u))}{j(2j+1)}, \qquad j \geq 1, \tag{2.6}$$

$$A_0(u) = 1, \qquad A_j(u) = \frac{1}{j}\sum_{h=1}^{j} h\ell_h(u)A_{j-h}(u). \tag{2.7}$$

Shifted Stirling expansion and Bernoulli reflection give

$$W_n(u) = x_n^{-1-2u}\left(\sum_{j=0}^{K-1} A_j(u)x_n^{-2j} + O(x_n^{-2K})\right). \tag{2.8}$$

For fixed K, the remainder is locally uniform in pole-free compact parameter sets. Every fixed parameter derivative is controlled on a slightly larger compact set by Cauchy's formula. Only even inverse powers occur in the correction series.

Proposition 2.1 (Prior centered subtraction and collapse)**.** *For $K \geq 1$ and $-K < \operatorname{Re} u < d_0$,*

$$\begin{aligned}\mathcal{R}_K(u) &= \sum_{n\geq 0}\left[W_n(u) - \sum_{j=0}^{K-1} A_j(u)x_n^{-1-2u-2j}\right] \\ &= \frac{1}{2}\Gamma(u)P(u) - \sum_{j=0}^{K-1} A_j(u)\zeta(1+2u+2j,\kappa).\end{aligned} \tag{2.9}$$

The bracketed series and all its fixed derivatives converge locally normally. If $N \geq 0$, then

$$A_N(-N) = \rho_N = \frac{(-1)^N}{N!}(d_0)_N(1-b)_N(1-c)_N, \tag{2.10}$$

$$\mathcal{R}_{N+1}(-N) = \frac{\rho_N}{2}\left[\psi(N+1) - \psi(d_0+N) - \psi(b) - \psi(c) + 2\psi(\kappa)\right]. \tag{2.11}$$

Here $(a)_N$ is the rising factorial. These formulas include $\rho_N = 0$.

Proof, recalled to specify the input. The compact-uniform remainder in (2.8) is summable on the stated strip. On $\operatorname{Re} u > 0$, the sum follows by (2.4); analytic continuation gives (2.9). Its holomorphic left side proves removal of all apparent poles on the right.

At $u = -N$, compare residues. The Gamma term has residue $\rho_N/2$; the only polar zeta term is $j = N$, with residue $A_N(-N)/2$. This proves (2.10), first for generic parameters and then polynomially. The cancellation of the finite parts is encoded by

$$(\partial_u - D)A_N(u) = \sum_{h=1}^{N} \frac{B_{2h}(\kappa)}{h}A_{N-h}(u), \qquad D = \partial_\kappa + \partial_b + \partial_c. \tag{2.12}$$

Indeed $Dd_0 = 0$, and $\partial_u - D$ acts on the four entries of (2.5) by $(-1, 0, 0, 0)$. Apply this operator to $\exp(\sum_{j\geq 1} \ell_j y^j) = \sum_{j\geq 0} A_j y^j$. The Bernoulli derivative identity gives (2.12).

The constant term of the Gamma expression is

$$\frac{\rho_N}{2}[\psi(N+1) - \psi(d_0+N) - \psi(b-N) - \psi(c-N)].$$

The polar zeta counterterm contributes $A'_N(-N)/2 - \rho_N \psi(\kappa)$. The lower zeta values are Bernoulli polynomials; (2.12) cancels them and replaces the coefficient derivative by $D\rho_N$. Finally,

$$D\rho_N = \rho_N \sum_{h=1}^{N} \left(\frac{1}{b-h} + \frac{1}{c-h} \right),$$

and the digamma recurrence produces (2.11). Continuity proves the exceptional cases. This is the centered continuation's proof mechanism [7]. □

3 A complete rationality and finite-termination classification

At a fixed integral resonance, put

$$d = d_0 + N, \qquad \alpha = (\kappa, b-\kappa, c-\kappa, d-\kappa), \qquad b+c+d = 1+2\kappa+N. \tag{3.1}$$

The meromorphic function whose sampled values are $W_n(-N)$ is

$$F_N(x) = x \prod_{i=1}^{4} \frac{\Gamma(x+\alpha_i)}{\Gamma(x+1-\alpha_i)}. \tag{3.2}$$

Here "rational" means belonging to $\mathbb{R}(x)$; it does not require rational coefficients or rational values at integer arguments. The three parameters b, c, d are symmetric in this expression, although $d_0 > 0$ singles out one of them in the original convergence domain.

3.1 Two general lemmas

Lemma 3.1 (Rational Gamma quotients)**.** *For real shifts $a_1, \ldots, a_r$ and $e_1, \ldots, e_r$, the quotient $\prod_i \Gamma(x+a_i)/\Gamma(x+e_i)$ is rational in x if and only if the two multisets of shifts agree modulo $\mathbb{Z}$.*

Proof. If the multisets agree, pair equal residue classes and use $\Gamma(z+h)/\Gamma(z) = (z)_h$ for integral h, interpreting negative h by the reciprocal product. This gives a rational function. Conversely, in each residue class of shifts modulo $\mathbb{Z}$, numerator Gamma factors contribute one pole at every sufficiently far negative point of that arithmetic progression. Denominator Gamma factors contribute the corresponding zeros. Unless the multiplicities agree, the quotient has infinitely many zeros or poles in that progression. A nonzero rational function cannot do so. Finite initial cancellations cannot change this conclusion. □

Lemma 3.2 (Formal termination is exact)**.** *Suppose a meromorphic function F has a nonzero finite Laurent asymptotic expansion P to all algebraic orders on the positive real axis, and $F(x+1) = R(x)F(x)$ for a rational function R. Then $F = P$, provided $F(x)/P(x) \to 1$ as $x \to +\infty$.*

Proof. For every prescribed power M, the difference $F(x) - P(x)$ is $O(x^{-M})$. Substitution into the difference equation implies that $P(x+1) - R(x)P(x)$ decays faster than every power. This is a rational function; therefore it vanishes identically. Away from the finite exceptional sets, F/P is consequently 1-periodic. For each sufficiently large real x,

$$\frac{F(x)}{P(x)} = \frac{F(x+j)}{P(x+j)} \longrightarrow 1 \qquad (j \to \infty).$$

Thus the quotient equals one on an interval, and meromorphic continuation proves the identity. This argument is what rules out an unexamined beyond-all-orders correction. □

For (3.2), the required rational difference equation is

$$\frac{F_N(x+1)}{F_N(x)} = \frac{x+1}{x}\prod_{i=1}^{4}\frac{x+\alpha_i}{x+1-\alpha_i}. \tag{3.3}$$

Its leading asymptotic power is x^{2N-1}.

3.2 The classification theorem

Theorem 3.3 (Rationality and termination)**.** *Assume* (2.1) *and* $N \geq 0$.

(i) *The function* $F_N(x)$ *is rational if and only if at least one of* b, c, d *is an integer.*

(ii) *Its formal centered inverse-power expansion terminates if and only if* F_N *is a Laurent polynomial. This occurs precisely in either of the following cases:*

$$\textit{some member of } \{b,c,d\} \textit{ lies in } \{1,\dots,N\}; \tag{3.4}$$

or, up to permutation,

$$(b,c,d) = (N+1,\kappa,\kappa). \tag{3.5}$$

In the first case F_N *is a polynomial. In the second case,*

$$F_N(x) = \frac{1}{x}\prod_{j=1}^{N}(x^2-(\kappa-j)^2). \tag{3.6}$$

The empty product at $N=0$ *is one. Overlap of the two cases is allowed.*

Proof. Since $\sum_i \alpha_i = N+1$, Lemma 3.1 says that rationality is equivalent to invariance of the four-element multiset $\{\alpha_i \bmod \mathbb{Z}\}$ under negation. Such an invariant multiset, with integral total sum, has a partner for α_1 among the other three elements, with integral pair sum. To see the only delicate case, suppose α_1 is a fixed class under negation, hence 0 or 1/2. If no second copy of that fixed class occurs, the other nonfixed classes pair off, leaving an odd number of the other fixed class. The total sum is then half-integral, a contradiction. The possible partner sums are exactly b, c, d.

Conversely, if one of those three pair sums is integral, the complementary pair sum is integral as well, because their total is $N+1$. Pairing proves negation invariance and hence rationality. This establishes (i).

If the formal expansion terminates, it is a finite Laurent polynomial with leading coefficient one. Equation (3.3) and Lemma 3.2 identify it exactly with F_N. The converse is immediate from uniqueness of the asymptotic expansion. It remains to classify the rational functions which are Laurent polynomials.

Choose an integer member m of $\{b,c,d\}$ and rename the other two c_*, d_*. Positivity implies $m \geq 1$. If $m \leq N$, Gamma recurrence gives

$$F_N(x) = x\prod_{j=1}^{m-1}(x^2-(\kappa-j)^2)\prod_{r=1}^{N-m}(x^2-(\kappa-c_*+r)^2). \tag{3.7}$$

This is the short-integer polynomial locus.

Suppose no integer member is at most N, and choose m to be the *smallest integer member*. Put $L = m - N \geq 1$. The same recurrence now gives the rational expression

$$F_N(x) = x \frac{\prod_{j=1}^{m-1} (x^2 - (\kappa - j)^2)}{\prod_{r=0}^{L-1} (x^2 - (\kappa - c_* - r)^2)}. \tag{3.8}$$

A numerator and denominator factor can cancel only if

$$c_* + r = j \quad \text{or} \quad d_* + L - 1 - r = j, \qquad 1 \leq j \leq m - 1,$$

where the second equality uses $c_* + d_* = 1 + 2\kappa - L$. If the parameter on the left is nonintegral, neither equality is possible. If integral, it is at least m by minimality of m, so neither equality is possible again. Thus no quadratic factors cancel.

A Laurent polynomial can have no pole away from $x = 0$. Consequently, every denominator root in (3.8) must be zero. The consecutive numbers $\kappa - c_* - r$, $0 \leq r < L$, can all vanish only when $L = 1$ and $c_* = \kappa$. The parameter sum then forces $d_* = \kappa$ and $m = N + 1$. Formula (3.6) follows. This proves necessity and sufficiency, including multiple roots and zero factors by meromorphic continuation. □

Remark 3.4 (Exact scope of the symmetric description). Under $d_0 > 0$, one has $d > N$. Thus the short-integer alternative is actually attained by b or c in the original parameter chart. The exceptional locus $d = N + 1$, $b = c = \kappa$, $d_0 = 1$ is admissible for every $\kappa > 0$. The permutations with $b = N + 1$ or $c = N + 1$ require $\kappa > N$. The symmetric theorem is useful, but the positivity domain must still be enforced after a permutation.

3.3 Consequences and examples

A terminating centered expression need not be a polynomial: the generic exceptional expression has a simple x^{-1} term. Its residue coefficient is

$$\rho_N = (-1)^N \prod_{j=1}^{N} (\kappa - j)^2 \quad \text{on } (b, c, d) = (\kappa, \kappa, N + 1). \tag{3.9}$$

This agrees with (2.10). On all terminating loci,

$$W_n(-N) = \sum_{j=0}^{N} A_j(-N) x_n^{2N-1-2j}, \qquad \mathcal{R}_{N+1}(-N) = 0, \tag{3.10}$$

with any unused final coefficients zero.

For example, $N = 2$, $\kappa = b = c = 1/2$, $d_0 = 1$ gives

$$F_2(x) = \frac{(x^2 - 1/4)(x^2 - 9/4)}{x} = x^3 - \frac{5}{2}x + \frac{9}{16x}.$$

By contrast, $N = 1$, $\kappa = 2$, $b = 3$, $c = 3/4$, $d_0 = 5/4$ is rational but does not terminate:

$$F_1(x) = \frac{x^3(x^2 - 1)}{(x^2 - 25/16)(x^2 - 1/16)}.$$

Both statements are identities of meromorphic functions; evaluating only at large integer indices is not needed for the classification.

4 Zero summed tails without rational summands

The implication from termwise termination to a zero sum has no converse. The collapsed expression in Proposition 2.1 permits a precise local construction rather than a numerical search for examples.

Theorem 4.1 (Nonrational zero-remainder curves)**.** *For every $N \geq 0$ and $0 < \kappa < 1$, there is a real analytic even function $s(v)$ near $v = 0$, with $s(0) = 0$, such that the parameters*

$$b = \kappa + s(v) + v, \quad c = \kappa + s(v) - v, \quad d_0 = 1 - 2s(v) \tag{4.1}$$

satisfy

$$\mathcal{R}_{N+1}(-N) = 0. \tag{4.2}$$

For all sufficiently small real $v \neq 0$, the corresponding F_N is not rational and its centered expansion does not terminate. Moreover,

$$s''(0) = \frac{\psi''(\kappa)}{\psi'(N+1) - \psi'(\kappa)} > 0. \tag{4.3}$$

Proof. Consider the real analytic function

$$\begin{aligned} B(s, v) = \psi(N+1) - \psi(N+1-2s) \\ - \psi(\kappa + s + v) - \psi(\kappa + s - v) + 2\psi(\kappa). \end{aligned}$$

At the origin, $B = 0$,

$$B_s = 2[\psi'(N+1) - \psi'(\kappa)] \neq 0, \qquad B_v = 0, \qquad B_{vv} = -2\psi''(\kappa).$$

The trigamma is strictly decreasing on the positive axis, since $\psi''(x) = -2\sum_{n\geq 0}(n+x)^{-3} < 0$. The analytic implicit function theorem gives a unique analytic $s(v)$ solving $B(s(v), v) = 0$. Evenness follows from uniqueness and $B(s, -v) = B(s, v)$. Differentiating twice gives (4.3); in particular $s(v) > 0$ for small nonzero real v.

All parameters remain positive. Shrink the neighborhood so that $b, c \in (0, 1)$ and $N < d = N + 1 - 2s(v) < N + 1$ for $v \neq 0$. No member of $\{b, c, d\}$ is then integral, and $\rho_N \neq 0$. Equation (2.11) proves (4.2), while Theorem 3.3 proves nonrationality and nontermination. The argument uses an exact implicit family, not the accuracy of a numerical zero. □

This theorem is not a correction to an asserted converse in the inspected report: no such converse was found. It is a new separation result and an audit guard. A small or zero total residual cannot be used to infer summand-level collapse.

5 A symmetric resolvent and its Stieltjes boundary

We now use $a > 0$ for the center and set $\kappa = a$, $b = a + v$, $c = a - v$, $d_0 = 1$. Let $|v| < a$ and define

$$T_n(u; a) = \frac{\Gamma(n + 2a)\Gamma(n + 1 - u)}{\Gamma(n + 1)\Gamma(n + 2a + u)}, \tag{5.1}$$

$$Q_a(u, v) = \frac{\pi u}{\sin \pi u} \frac{\Gamma(a + v)\Gamma(a - v)}{\Gamma(a + v + u)\Gamma(a - v + u)}. \tag{5.2}$$

Then $W_n(u) = x_n T_n(u; a)/(x_n^2 - v^2)$, where $x_n = n + a$. The symmetry makes the resolvent denominator appear without any approximation.

Theorem 5.1 (Ordinary convergent resolvent identity)**.** *For $a > 0$, $|v| < a$, and $-1 < \operatorname{Re} u < 1$,*

$$\sum_{n\geq 0} \frac{v^2 T_n(u;a)}{x_n(x_n^2 - v^2)} = \frac{Q_a(u,v) - Q_a(u,0)}{2u}. \tag{5.3}$$

The right side is interpreted removably at $u = 0$. The series and all fixed u- and v-derivatives converge locally normally.

Proof. In $0 < \operatorname{Re} u < 1$, subtract two instances of (2.4), one with v and one with zero. The elementary identity

$$\frac{x}{x^2 - v^2} - \frac{1}{x} = \frac{v^2}{x(x^2 - v^2)}$$

and $\Gamma(1+u)\Gamma(1-u) = \pi u/\sin \pi u$ give (5.3).

The centered expansion gives $T_n(u;a) = x_n^{-2u}(1 + O(x_n^{-2}))$ locally uniformly. Thus the summand is $O(n^{-3-2\operatorname{Re} u})$ on compact sets of the stated domain. Cauchy's formula on slightly larger compact sets gives normal convergence of each fixed derivative. The factors $1/\Gamma(n+2a+u)$ are entire and introduce no poles; the numerator Gamma factors are pole-free for $\operatorname{Re} u < 1$. Both sides therefore continue to the full strip. Since $Q_a(0,v) = 1$, the apparent pole at zero is removable. □

The subtraction suppresses the only divergent moment. To see exactly what is removed, retain that moment and use

$$\zeta(1+s,a) = \frac{1}{s} + \sum_{m\geq 0} \frac{(-1)^m \gamma_m(a)}{m!} s^m. \tag{5.4}$$

Theorem 5.2 (All Stieltjes boundary jets)**.** *On the same domain as Theorem 5.1,*

$$\begin{aligned} \mathcal{B}_a(u,v) &= \sum_{n\geq 0} \left[\frac{x_n T_n(u;a)}{x_n^2 - v^2} - x_n^{-1-2u} \right] \\ &= \frac{Q_a(u,v)}{2u} - \zeta(1+2u,a). \end{aligned} \tag{5.5}$$

The bracketed series is locally normally convergent, and for every integer $m \geq 0$,

$$\partial_u^m \mathcal{B}_a(0,v) = \frac{\partial_u^{m+1} Q_a(0,v)}{2(m+1)} - (-2)^m \gamma_m(a). \tag{5.6}$$

Subtracting the $v = 0$ identity cancels $\gamma_m(a)$ at every order.

Proof. The leading centered term of the bracket is canceled, leaving the same summable order as in Theorem 5.1. The identity follows first in the convergent strip and then by continuation. Expand both terms on the right at zero. The common coefficient $1/(2u)$ cancels; the coefficient of u^m, multiplied by $m!$, is (5.6). Termwise differentiation was established before the expansion. □

Write $\psi_\pm = \psi(a \pm v)$ and $\psi'_\pm = \psi'(a \pm v)$. The first two boundary values are

$$\mathcal{B}_a(0,v) = \psi(a) - \frac{\psi_+ + \psi_-}{2}, \tag{5.7}$$

$$\partial_u \mathcal{B}_a(0,v) = \frac{(\psi_+ + \psi_-)^2 - \psi'_+ - \psi'_- + 2\zeta(2)}{4} + 2\gamma_1(a). \tag{5.8}$$

At $v = 0$, the second formula becomes the ordinary convergent identity

$$\sum_{n\geq 0} \frac{2\log(n+a) - \psi(n+1) - \psi(n+2a)}{n+a} = \psi(a)^2 - \frac{1}{2}\psi'(a) + \frac{1}{2}\zeta(2) + 2\gamma_1(a). \tag{5.9}$$

For complex a in a pole-free continuation, the square is algebraic, not an absolute square. The numerical implementation in this package uses positive real a for Stieltjes constants.

6 Finite polygamma formulas for every higher resolvent moment

For $k \geq 1$, expansion of (5.3) in v gives

$$\sum_{n\geq 0} \frac{T_n(u;a)}{(n+a)^{2k+1}} = \frac{[v^{2k}]Q_a(u,v)}{2u}. \tag{6.1}$$

This is already a finite Gamma-derivative expression. An explicit coefficient algebra makes its all-orders content transparent. Put

$$D_j(u) = \frac{2}{(2j)!}[\psi^{(2j-1)}(a) - \psi^{(2j-1)}(a+u)]. \tag{6.2}$$

Taylor expansion in v yields

$$\frac{[v^{2k}]Q_a(u,v)}{2u} = \frac{Q_a(u,0)}{2u}[z^k]\exp\left(\sum_{j\geq 1} D_j(u)z^j\right). \tag{6.3}$$

Only $j \leq k$ are needed. The numerator vanishes at $u = 0$ before division, because every $D_j(0)$ is zero.

Let Y_m be the complete exponential Bell polynomial, defined by

$$\exp\left(\sum_{r\geq 1} g_r \frac{u^r}{r!}\right) = \sum_{m\geq 0} Y_m(g_1,\ldots,g_m)\frac{u^m}{m!}.$$

For the left side of (6.1), the Bell data are

$$g_{n,r} = (-1)^r\psi^{(r-1)}(n+1) - \psi^{(r-1)}(n+2a). \tag{6.4}$$

Consequently every sum

$$\sum_{n\geq 0} \frac{Y_m(g_{n,1},\ldots,g_{n,m})}{(n+a)^{2k+1}}, \qquad m \geq 0,\ k \geq 1, \tag{6.5}$$

is given by the mth derivative of the finite expression (6.3). All its constants are polygamma values at a and ordinary even zeta values. This does not assert a reduction of arbitrary unrelated polygamma products; it evaluates this particular Bell family.

Corollary 6.1 (An all-weight linear identity)**.** *For $a > 0$ and every integer $k \geq 1$,*

$$\sum_{n\geq 0} \frac{\psi(n+1)+\psi(n+2a)}{(n+a)^{2k+1}}$$
$$= \frac{2k+1}{2}\zeta(2k+2,a) + 2\psi(a)\zeta(2k+1,a) - \sum_{j=1}^{k-1}\zeta(2j+1,a)\zeta(2k-2j+1,a). \tag{6.6}$$

Proof. The required sum is the negative first u-derivative of (6.1), since $\partial_u T_n(0;a) = -\psi(n+1) - \psi(n+2a)$. In (6.3), retain terms through degree two in u:

$$D_j(u) = -\frac{2\psi^{(2j)}(a)}{(2j)!}u - \frac{\psi^{(2j+1)}(a)}{(2j)!}u^2 + O(u^3), \quad Q_a(u,0) = 1 - 2\psi(a)u + O(u^2).$$

The exponential contributes a linear term and the quadratic convolution of two D_j's. Insert $\psi^{(r)}(a) = (-1)^{r+1}r!\zeta(r+1,a)$ for $r \geq 1$. Coefficient extraction gives (6.6). □

Corollary 6.2 (A quadratic identity at every positive center)**.** *For $a > 0$,*

$$\sum_{n\geq 0} \frac{[\psi(n+1)+\psi(n+2a)]^2+\psi'(n+1)-\psi'(n+2a)}{(n+a)^3} = -\frac{1}{6}\psi^{(4)}(a)+\psi(a)\psi^{(3)}(a)-[2\psi(a)^2+\zeta(2)-\psi'(a)]\psi''(a). \tag{6.7}$$

Proof. For $k=1$, the right side of (6.3) is $Q_a(u,0)[\psi'(a)-\psi'(a+u)]/(2u)$. Its second derivative at zero is the displayed expression, using

$$Q_a(u,0) = 1-2\psi(a)u+[2\psi(a)^2+\zeta(2)-\psi'(a)]u^2+O(u^3).$$

The second derivative of T_n is exactly the numerator in the summand. Normal convergence justifies the operation. □

7 A Gamma generator for nonlinear harmonic sums

The half-center $a=1/2$ removes the shifted polygamma arguments after one normalization. Define

$$P_n(u) = \frac{(1-u)_n}{(1+u)_n} = \prod_{j=1}^{n}\frac{j-u}{j+u}, \qquad P_0(u)=1, \tag{7.1}$$

and

$$\mathcal{G}(u,t) = \frac{\Gamma(1+u)^2\Gamma((1+t)/2)\Gamma((1-t)/2)}{\Gamma((1+t)/2+u)\Gamma((1-t)/2+u)}. \tag{7.2}$$

Theorem 7.1 (Normalized half-center generator)**.** *For $|t|<1$ and $\operatorname{Re} u > -1$,*

$$\sum_{n\geq 0}\frac{t^2P_n(u)}{(2n+1)((2n+1)^2-t^2)} = \frac{\mathcal{G}(u,t)-\mathcal{G}(u,0)}{4u}. \tag{7.3}$$

The singularity at $u=0$ is removable. In particular, for every $k\geq 1$,

$$\sum_{n\geq 0}\frac{P_n(u)}{(2n+1)^{2k+1}} = \frac{[t^{2k}]\mathcal{G}(u,t)}{4u}. \tag{7.4}$$

Every fixed Taylor coefficient at $u=0$ can be summed termwise.

Proof. Set $a=1/2$ and $v=t/2$ in (5.3), and multiply by $\Gamma(1+u)/\Gamma(1-u)$. Then T_n becomes P_n and Q_a becomes $\mathcal{G}$. The rational summand acquires a factor two when $x_n=(2n+1)/2$ is substituted; this is the source of $4u$ rather than $2u$ on the right of (7.3).

Initially the calculation applies on $-1<\operatorname{Re} u<1$ away from removable factors. The finite product (7.1) is holomorphic for $\operatorname{Re} u>-1$. On compact subsets, the Gamma-ratio form gives $P_n(u)=O(n^{-2\operatorname{Re} u})$, uniformly also across positive integral u by using the entire reciprocal-Gamma factor. The sum in (7.3) is therefore locally normally convergent on the larger half-plane. The Gamma expression is holomorphic there apart from its removable $u=0$ singularity. Analytic continuation extends the identity. Expansion in t and Cauchy's formula prove (7.4) and its coefficient assertions. □

7.1 Weighted compositions of generalized harmonic numbers

For a composition $\mathbf{r} = (r_1, \ldots, r_d)$, define

$$H_n(\mathbf{r}) = \sum_{n \geq j_1 > \cdots > j_d \geq 1} \frac{1}{j_1^{r_1} \cdots j_d^{r_d}}, \qquad H_n(\varnothing) = 1. \tag{7.5}$$

The notation $\mathbf{r} \vDash m$ means that all $r_i \geq 1$ and $\sum_i r_i = m$. For $m = 0$ there is the single empty composition. Put

$$C_m(n) = \sum_{\mathbf{r} \vDash m} 2^{\ell(\mathbf{r})} H_n(\mathbf{r}). \tag{7.6}$$

Lemma 7.2 (Finite composition identity)**.** *For every $n \geq 0$,*

$$P_n(u) = \sum_{m \geq 0} (-1)^m C_m(n) u^m \tag{7.7}$$

as a Taylor series near zero.

Proof. Each finite factor is $(1 - u/j)/(1 + u/j) = 1 + 2\sum_{r \geq 1} (-1)^r (u/j)^r$. Choose the non-constant terms at a strictly decreasing collection of indices and sum their positive exponents. The resulting coefficient is exactly (7.6). There is no infinite summation over n in this finite identity. □

For $H_n = H_n^{(1)}$ and $H_n^{(r)} = \sum_{j=1}^n j^{-r}$,

$$\begin{aligned} &C_0(n) = 1, && C_1(n) = 2H_n, && C_2(n) = 2H_n^2, \\ &C_3(n) = \frac{4H_n^3 + 2H_n^{(3)}}{3}, && C_4(n) = \frac{2H_n^4 + 4H_n H_n^{(3)}}{3}. \end{aligned} \tag{7.8}$$

These also follow from

$$\log P_n(u) = -2 \sum_{j \geq 0} \frac{H_n^{(2j+1)}}{2j+1} u^{2j+1}. \tag{7.9}$$

Thus only odd-index harmonic numbers enter the logarithm, although the equivalent composition expression runs over all compositions.

Theorem 7.3 (All-weight harmonic composition sum formula)**.** *For every $m \geq 0$ and $k \geq 1$,*

$$\boxed{\sum_{n \geq 0} \frac{C_m(n)}{(2n+1)^{2k+1}} = \frac{(-1)^m}{4} [u^{m+1} t^{2k}] \mathcal{G}(u,t).} \tag{7.10}$$

Every value belongs to $\mathbb{Q}[\log 2, \zeta(2), \zeta(3), \ldots]$. An explicit finite expression is obtained by exponentiating

$$\begin{aligned} \log \mathcal{G}(u,t) = 4u \log 2 &+ \sum_{r \geq 2} \frac{2(-1)^r (2 - 2^r)}{r} \zeta(r) u^r \\ &+ \sum_{\substack{r \geq 1 \\ j \geq 1}} \frac{2(-1)^{r+1}}{r} \binom{r + 2j - 1}{2j} (2^r - 2^{-2j}) \zeta(r + 2j) u^r t^{2j}. \end{aligned} \tag{7.11}$$

Only terms of u-degree at most $m + 1$ and t-degree at most $2k$ are required. The resulting expression is homogeneous of weight $m + 2k + 1$ when $\log 2$ has weight one and $\zeta(r)$ has weight r.

Proof. Combine (7.4) and Lemma 7.2; normal convergence permits coefficient extraction. Equivalently, $C_m(n)$ is bounded by a constant times $(1+\log n)^m$ for fixed m, so the displayed ordinary sum converges absolutely.

For completeness, expand the logarithm of (7.2) in u. Its coefficient of u^r is

$$\frac{2\psi^{(r-1)}(1) - \psi^{(r-1)}((1+t)/2) - \psi^{(r-1)}((1-t)/2)}{r!}.$$

The $r=1$, $t=0$ coefficient is $4\log 2$. At $t=0$ and $r\geq 2$, use $\psi^{(r-1)}(1) = (-1)^r(r-1)!\zeta(r)$ and $\psi^{(r-1)}(1/2) = (-1)^r(r-1)!(2^r-1)\zeta(r)$. For the coefficient of t^{2j}, expand the symmetric pair and use the same half-argument formula at order $r+2j-1$. Simplification gives (7.11).

Every monomial in the logarithm has constant weight equal to its u-degree plus its t-degree. Exponentiation preserves this grading. The coefficient in (7.10) therefore has the asserted weight. These are formal grading statements; they do not assert arithmetic independence of the constants. □

7.2 Explicit linear, quadratic, cubic, and quartic identities

Write $L = \log 2$. The first four nontrivial rows of (7.10), at $k=1$, simplify to

$$\sum_{n\geq 0} \frac{H_n}{(2n+1)^3} = \frac{45}{32}\zeta(4) - \frac{7}{4}L\zeta(3), \tag{7.12}$$

$$\sum_{n\geq 0} \frac{H_n^2}{(2n+1)^3} = \frac{31}{8}\zeta(5) - \frac{45}{8}L\zeta(4) + \frac{7}{2}L^2\zeta(3) - \frac{7}{8}\zeta(2)\zeta(3), \tag{7.13}$$

$$\begin{aligned}\sum_{n\geq 0} \frac{2H_n^3 + H_n^{(3)}}{(2n+1)^3} &= -14L^3\zeta(3) + \frac{21}{2}L\zeta(2)\zeta(3) - \frac{21}{4}\zeta(3)^2 \\ &\quad + \frac{135}{4}L^2\zeta(4) - \frac{93}{2}L\zeta(5) + \frac{945}{64}\zeta(6),\end{aligned} \tag{7.14}$$

$$\begin{aligned}\sum_{n\geq 0} \frac{H_n^4 + 2H_nH_n^{(3)}}{(2n+1)^3} &= 14L^4\zeta(3) - 21L^2\zeta(2)\zeta(3) + 21L\zeta(3)^2 \\ &\quad - 45L^3\zeta(4) + 93L^2\zeta(5) - \frac{945}{16}L\zeta(6) \\ &\quad - \frac{39}{2}\zeta(3)\zeta(4) - \frac{93}{4}\zeta(2)\zeta(5) + \frac{1143}{16}\zeta(7).\end{aligned} \tag{7.15}$$

The proof of each is finite coefficient extraction from a proved analytic identity. The exact script performs that extraction in a polynomial algebra and checks these simplified presentations using only the even-zeta relations. The displayed combinations, not H_n^3 and H_n^4 separately, are what the theorem evaluates. Nor are these sums the manuscript's alternating Gaussian sums S_p.

8 All-depth polylogarithms, integrals, and primitives

We use decreasing-index multiple polylogarithms

$$\mathrm{Li}_{r_1,\dots,r_d}(z_1,\dots,z_d) = \sum_{j_1>\cdots>j_d\geq 1} \frac{z_1^{j_1}\cdots z_d^{j_d}}{j_1^{r_1}\cdots j_d^{r_d}}. \tag{8.1}$$

The single-variable abbreviation $\mathrm{Li}_{\mathbf{r}}(z) = \mathrm{Li}_{\mathbf{r}}(z,1,\dots,1)$ is used below. For $m\geq 1$, set

$$\mathfrak{L}_m(z) = \sum_{\mathbf{r}\vDash m} 2^{\ell(\mathbf{r})}\,\mathrm{Li}_{\mathbf{r}}(z), \qquad \mathfrak{L}_0(z) = 1. \tag{8.2}$$

The finite-harmonic generating function is

$$\sum_{n\geq 0} C_m(n) z^n = \frac{\mathfrak{L}_m(z)}{1-z}, \qquad |z| < 1. \tag{8.3}$$

Indeed, for each fixed decreasing tuple, the allowed upper bound n runs from its largest member to infinity.

Theorem 8.1 (Integral and colored-polylogarithm forms)**.** *For integers $m \geq 0$, $p \geq 1$, and $0 < z < 1$, put*

$$F_{m,p}(z) = \sum_{n\geq 0} \frac{C_m(n) z^{2n+1}}{(2n+1)^p}. \tag{8.4}$$

Then

$$F_{m,p}(z) = \frac{1}{(p-1)!} \int_0^z \log^{p-1}(z/t) \frac{\mathfrak{L}_m(t^2)}{1-t^2}\, \mathrm{d}t, \tag{8.5}$$

$$F_{m,p}(z) = 2^{m-1} \sum_{\mathbf{r}\vDash m} \sum_{\epsilon_0,\ldots,\epsilon_d \in \{\pm 1\}} \epsilon_0 \operatorname{Li}_{p,r_1,\ldots,r_d}(\epsilon_0 z, \epsilon_1, \ldots, \epsilon_d), \tag{8.6}$$

where $d = \ell(\mathbf{r})$. Formula (8.6) extends to $|z| < 1$ by analyticity. For $p \geq 2$, both formulas extend to $z = 1$ by ordinary absolute convergence. In particular, for $p = 2k+1$, $k \geq 1$, Theorem 7.3 evaluates their endpoints in ordinary zeta values and $\log 2$ at every depth.

Proof. For each n,

$$\frac{1}{(p-1)!} \int_0^z t^{2n} \log^{p-1}(z/t)\, \mathrm{d}t = \frac{z^{2n+1}}{(2n+1)^p}.$$

Insert (8.3). Positivity for real z, or absolute convergence inside the unit disk, permits the interchange.

For the colored expression, fix a composition of length d. Replace the inner harmonic indices j_i by even integers $h_i = 2j_i$ and the outer denominator $2n+1$ by h_0. This contributes the factor 2^m from the denominators. The inequalities become $h_0 > h_1 > \cdots > h_d \geq 1$, with h_0 odd and the others even. The parity filters are

$$\mathbf{1}_{h_0 \text{ odd}} = \frac{1}{2} \sum_{\epsilon_0 = \pm 1} \epsilon_0 \epsilon_0^{h_0}, \qquad \mathbf{1}_{h_i \text{ even}} = \frac{1}{2} \sum_{\epsilon_i = \pm 1} \epsilon_i^{h_i}.$$

Multiplication by the composition weight 2^d leaves $2^m 2^d / 2^{d+1} = 2^{m-1}$, proving (8.6). At $m = 0$, the convention gives the familiar $(\operatorname{Li}_p(z) - \operatorname{Li}_p(-z))/2$.

For fixed m, the coefficients grow at most logarithmically to power m. Thus $p \geq 2$ makes the endpoint sum absolutely convergent. The corresponding multiple sums have first index at least two, so every colored term is also absolutely convergent. This permits separate endpoint limits; no regularization is being hidden here. □

Corollary 8.2 (An exact dilogarithmic integral)**.** *One has*

$$\begin{aligned} &\frac{1}{2} \int_0^1 \frac{\log^2 x}{1-x^2} [\operatorname{Li}_2(x^2) + \log^2(1-x^2)]\, \mathrm{d}x \\ &\quad = \frac{31}{8}\zeta(5) - \frac{45}{8} \log 2\, \zeta(4) + \frac{7}{2} \log^2 2\, \zeta(3) - \frac{7}{8} \zeta(2)\zeta(3). \end{aligned} \tag{8.7}$$

Proof. The identity

$$\sum_{n\geq 0} H_n^2 z^n = \frac{\operatorname{Li}_2(z) + \log^2(1-z)}{1-z}$$

follows by splitting the double harmonic sum into its diagonal and two strict triangles (equivalently, $\operatorname{Li}_{1,1}(z) = \frac{1}{2}\log^2(1-z)$). Use the integral in (8.5) at $p = 3$, then (7.13). The endpoint logarithms are integrable; the preceding positive-sum argument also proves convergence directly. □

8.1 Euler primitives and a Gamma primitive

Set

$$F_{m,0}(z) = \frac{z\mathfrak{L}_m(z^2)}{1-z^2}. \tag{8.8}$$

Termwise differentiation inside the disk gives the complete Euler primitive ladder

$$z\frac{\mathrm{d}}{\mathrm{d}z}F_{m,p}(z) = F_{m,p-1}(z), \qquad F_{m,p}(z) = \int_0^z \frac{F_{m,p-1}(t)}{t}\,\mathrm{d}t. \tag{8.9}$$

The basepoint zero fixes the integration constants. Formula (8.6) supplies explicit primitives in multiple polylogarithms, and the odd-denominator endpoint values for $p \geq 3$ are supplied by (7.10). The endpoint $p = 1$ must not be assigned a finite value by this argument.

At $u = 0$, the general resolvent also has an elementary Gamma primitive:

$$\sum_{n\geq 0}\left[\operatorname{arctanh}\frac{v}{n+a} - \frac{v}{n+a}\right] = v\psi(a) - \frac{1}{2}\log\frac{\Gamma(a+v)}{\Gamma(a-v)}, \qquad |v| < a. \tag{8.10}$$

Both sides vanish at zero. Their derivatives are the two sides of (5.3) at $u = 0$, by (5.7). The logarithm is the analytic branch normalized at $v = 0$; for real $|v| < a$ it is the real logarithm. Normal convergence follows from the cubic cancellation in the summand.

8.2 Stieltjes primitives with a fixed convention

Hurwitz spectral derivatives provide a compatible primitive for each boundary constant. Differentiating $\partial_a\zeta(s,a) = -s\zeta(s+1,a)$ at $s = 0$ using (5.4) yields

$$\partial_a\zeta^{(m+1)}(0,a) = (-1)^{m+1}(m+1)\gamma_m(a). \tag{8.11}$$

Consequently, for $a, a_0 > 0$,

$$\int_{a_0}^{a}\gamma_m(t)\,\mathrm{d}t = \frac{(-1)^{m+1}}{m+1}[\zeta^{(m+1)}(0,a) - \zeta^{(m+1)}(0,a_0)]. \tag{8.12}$$

The superscript on ζ differentiates the spectral variable. At $m = 0$, Lerch's identity gives the expected $-\log\Gamma(a) + \log\Gamma(a_0)$. These primitive identities are classical consequences of Hurwitz-zeta calculus, included to tie the boundary jets to the manuscript's integration conventions, not as a claim of new priority.

9 Zero-safe differentiation at terminating parameter values

A family may terminate at a parameter value while its derivative has an infinite tail. The normalized half-center makes this explicit without special-function ambiguities.

Proposition 9.1 (Revived derivatives of a terminating product)**.** *For integers $M \geq 1$ and $n \geq M$,*

$$P_n(M) = 0, \qquad P_n'(M) = (-1)^M\frac{M!(M-1)!(n-M)!}{(n+M)!}. \tag{9.1}$$

For $0 \leq n < M$,

$$\begin{aligned} P_n(M) &= (-1)^n\frac{(M-1)!M!}{(M-1-n)!(M+n)!}, \\ P_n'(M) &= P_n(M)[H_{M-1} - H_{M-1-n} - H_{M+n} + H_M]. \end{aligned} \tag{9.2}$$

Proof. For $n \geq M$, differentiate the unique zero factor $(M-u)/(M+u)$ in the finite product (7.1); all other factors can then be evaluated at M. Their factorial products give (9.1). For $n < M$, no factor vanishes. Logarithmic differentiation of the finite nonzero product gives (9.2). No logarithm of a zero factor is taken. □

For $M = 1$, all $n \geq 1$ have $P_n'(1) = -1/(n(n+1))$. One exact family appearing in the differentiated generator is

$$\sum_{n\geq 1} \frac{1}{n(n+1)(2n+1)^{2k+1}} = 4k + 3 - 4\log 2 - 4\sum_{j=1}^{k}(1 - 2^{-2j-1})\zeta(2j+1), \quad k \geq 1. \tag{9.3}$$

It also has an independent elementary proof. With $x = 2n+1$, repeatedly use

$$\frac{1}{n(n+1)x^{2k+1}} = \frac{1}{n(n+1)x^{2k-1}} - \frac{4}{x^{2k+1}}.$$

The initial sum at $k = 0$ is $3 - 4\log 2$, obtained from $1/[n(n+1)(2n+1)] = 1/n + 1/(n+1) - 4/(2n+1)$ by partial sums. This proves (9.3).

The general warning about reciprocal-Gamma zeros was already explicit in the centered continuation [7]. The present formulas are a concrete finite-product realization, not a claim that the warning itself is new.

10 Verification, audit conclusions, and integration

10.1 What the executable checks establish

The package contains a finite exact coefficient engine and separate high-precision diagnostics. The exact tests use rational residue classes of Gamma shifts to construct divisors independently of the theorem's parameter classification. They compare rationality and Laurent termination on a grid of 10,356 valid parameter choices. The grid uses $0 \leq N \leq 5$, quarter-integer positive centers through 3, and quarter-integer positive b, c subject to $d_0 > 0$.

The full exact run executes 21,009 assertions: 10,356 rationality checks, 10,356 termination checks, 204 monomial-weight checks, four printed harmonic-identity checks, 56 finite composition checks, 30 revived-zero derivative checks, and three unweighted odd-denominator checks. These finite calculations are regression tests of formulas and implementations. They do not replace the all-parameter proofs.

The numerical checks compare independently summed series against Gamma, polygamma, zeta, and Stieltjes expressions. They include an independent quadrature of (8.7). Series are evaluated by a finite head plus an Euler–Maclaurin tail at 40 decimal digits. The assertion threshold is a scaled residual of 10^{-28}; it is not a rigorous error bound. All 28 diagnostics passed; the largest observed scaled residual was 5.511×10^{-40}. Full parameters, values, observed residuals, software versions, and methods appear in the result files. No interval certification or proof-assistant verification is claimed.

10.2 Corrections and guards

No false theorem was identified in the selected source ranges. The following distinctions are necessary for correct integration.

1. The canonical first-resonance proof and the preserved all-resonance proof have different editorial status. The present package does not relabel the already proved all-resonance formula as a newly settled conjecture.

2. A zero summed remainder is not finite termination. Theorem 4.1 proves this within the same family.
3. Formal termination needs an argument ruling out a flat remainder. Lemma 3.2 supplies that argument, rather than assuming that a truncated asymptotic expansion equals the function.
4. Reciprocal-Gamma zeros are zeros of values, not automatically of jets. Keep the entire factors or use finite products before differentiating.
5. The normalization constants are $\Gamma(u)P(u)/2$, $[v^{2k}]Q_a(u,v)/(2u)$, and $[t^{2k}]\mathcal{G}(u,t)/(4u)$. The Stieltjes expansion uses $2u$, not u.
6. The all-depth theorem evaluates a weighted sum over compositions and colors. It asserts neither a reduction of every individual colored multiple zeta value nor a minimality or independence theorem.

The suggested placement is an additive continuation of the centered Dougall report, with the classification linked from the canonical mixed-Dougall section. The harmonic/polylogarithmic sections can also be cross-referenced from the depth and integration chapters. The supplied integration note gives a compact insertion with distinct labels. Existing source files should not be silently replaced by this draft.

11 Further research questions

1. Interlaced resonances. Classify rationality and finite termination at $u = -N - 1/2$ in the same positive-parameter family. The shift multiset then has half-integral total sum, so the fixed-class argument in Theorem 3.3 changes. A full analysis should preserve initial reciprocal-Gamma zeros and distinguish polynomial from Laurent termination.

2. Larger Gamma quotients. For $x\prod_{i=1}^{2r}\Gamma(x+\alpha_i)/\Gamma(x+1-\alpha_i)$, can rationality and Laurent termination be described by an effective matching theorem and a finite arrangement of parameter loci? Modulo-integer matching decides rationality, but a comparably short cancellation-free classification may need additional structure beyond four factors.

3. Global geometry of exact zero remainders. The implicit curves in Theorem 4.1 are local. Determine the connected components and possible singularities of the zero set of the collapsed digamma bracket inside the positive domain. Which components meet the terminating locus, and which contain no rational points in the Gamma-quotient sense?

4. Independent harmonic coordinates outside the Bell family. At each weight, which individual sums such as $\sum H_n^3/(2n+1)^3$ or $\sum H_n^{(3)}/(2n+1)^3$ reduce to the single-zeta/logarithm algebra? The present theorem evaluates a specific combination and does not separate its components. A precise comparison algebra is needed before a reduction or obstruction can be claimed.

5. General rational centers. For rational a, the finite formula (6.3) lies in a polygamma algebra at a. Transport this formula to a conductor-explicit Dirichlet-L and cyclotomic-polylogarithm basis, with zero-safe treatment of principal-character poles. This is a reduction problem, not an assertion of independence of the resulting coordinates.

6. Twisted and alternating resolvents. Determine which numerator twists ω^n admit a comparably short Gamma evaluation, or a finite combination of such evaluations. An alternating analogue would bear more directly on the manuscript's Gaussian sums, but the nonalternating identities here do not imply it.

7. Relations to known multiple-zeta sum formulas. Give a complete comparison of (8.6) at $z = 1$ with published height, depth, and level-two sum formulas, including the hypergeometric harmonic-differentiation literature. This is an important priority and structural-comparison task left open by the limited source review, not evidence that such antecedents are absent.

8. Beyond the first Euler endpoint. Can additional parameter symmetries evaluate the even-p endpoints of $F_{m,p}(1)$ in a comparably explicit algebra? The integral and colored forms exist for every $p \geq 2$, but the Gamma generator here singles out odd $p \geq 3$.

9. Formal verification of the finite interfaces. Formalize Gamma-shift divisor matching, the four-class parity argument, the cancellation-free product classification, and the finite composition identity. These finite interfaces are natural preliminary targets before formalizing normal convergence and the Rogers–Dougall continuation.

A A finite coefficient algorithm

Let $T = t^2$ and write $\log \mathcal{G}(u,t) = \sum_{r\geq 1} L_r(T)u^r$. For a requested $[u^{m+1}T^k]$ coefficient, discard powers of T above k from every L_r, and retain only $r \leq m+1$. Define

$$G_0(T) = 1, \qquad G_n(T) = \frac{1}{n}\sum_{r=1}^{n} rL_r(T)G_{n-r}(T) \pmod{T^{k+1}}. \tag{A.1}$$

Then $G_n(T) = [u^n]\mathcal{G}(u,t) \pmod{T^{k+1}}$. The recurrence follows by differentiating an exponential formal series and comparing coefficients. It uses only finitely many rational operations. In particular, (7.10) is $(-1)^m[T^k]G_{m+1}(T)/4$.

The executable implementation uses the formal symbols `L` for $\log 2$, `Zr` for $\zeta(r)$, and `T` for t^2. The delivered coefficient table covers $0 \leq m \leq 5$, $1 \leq k \leq 3$. Even zeta values are simplified only when explicitly requested, using

$$\zeta(2j) = \frac{(-1)^{j+1}B_{2j}\,2^{2j}6^j}{2(2j)!}\zeta(2)^j. \tag{A.2}$$

No integer-relation algorithm or floating-point fitting is used.

B Domains and conventions at a glance

Identity or object	Domain and convention
Classification	$N \geq 0$, $\kappa, b, c, d_0 > 0$, $d_0 = 1 + 2\kappa - b - c$. Gamma denominators are entire reciprocal-Gamma factors.
General resolvent	$a > 0$, $\lvert v\rvert < a$, $-1 < \operatorname{Re} u < 1$; $u = 0$ is removable.
Half-center generator	$\lvert t\rvert < 1$, $\operatorname{Re} u > -1$; Taylor coefficients are ordinary coefficients, not derivatives.

Identity or object	Domain and convention
Harmonic sum formula	Integers $m \geq 0$, $k \geq 1$; all sums are ordinary and absolutely convergent.
Polylogarithm endpoint	$p \geq 2$; strictly decreasing nested indices. The $p = 1$ endpoint is not evaluated by this theorem.
Stieltjes constants	$\zeta(1+s,a) = s^{-1} + \sum(-1)^m \gamma_m(a) s^m/m!$; $\gamma_0(a) = -\psi(a)$.
Gamma primitive	$\lvert v \rvert < a$, normalized analytic logarithm at $v = 0$.
Numerical evidence	40-digit floating-point diagnostics; no certified error intervals and no formal proof-assistant claim.

References

[1] NIST Digital Library of Mathematical Functions, §16.4, *Argument Unity*, especially Eq. 16.4.9, Rogers–Dougall summation. https://dlmf.nist.gov/16.4. Accessed 11 October 2026.

[2] NIST Digital Library of Mathematical Functions, §5.5, *Functional Relations*, and §5.11, *Asymptotic Expansions*. https://dlmf.nist.gov/5.5; https://dlmf.nist.gov/5.11. Accessed 11 October 2026.

[3] NIST Digital Library of Mathematical Functions, §25.11, *Hurwitz Zeta Function*, and §25.12, *Polylogarithms*. https://dlmf.nist.gov/25.11; https://dlmf.nist.gov/25.12. Accessed 11 October 2026.

[4] Ce Xu and Weiping Wang, *Dirichlet type extensions of Euler sums*, Comptes Rendus Mathématique **361** (2023), 979–1010. DOI: 10.5802/crmath.453. The publisher abstract and bibliographic information were inspected; a full-paper priority comparison was not completed.

[5] Vladimir Reshetnikov, *ProveIt: Polylogarithms manuscript*, repository snapshot `da2792fc9cc8ef19c6a314be6c65e1437fbdc66b`. https://github.com/VladimirReshetnikov/ProveIt. Inspected files include `polylogarithms.tex` and selected chapters under `Analysis/Polylogarithms/docs/manuscript/`.

[6] ProveIt, *Mixed Gamma derivatives and exact harmonic tail identities*, canonical manuscript file `chapters/07-mixed-dougall-identities.tex`, lines 1–240, at the snapshot in [5]. The first-resonance and mixed-derivative results are antecedents here.

[7] ProveIt, *Centered Dougall Resonance: Exact Stieltjes Jets, Harmonic Identities, Polylogarithmic Subtraction, and Zeta Antiderivatives*, 10 October 2026. Preserved continuation under `Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/centered-dougall-resonance/`. Inspected: README; complete `integration/centered-dougall-resonance.tex`; `article.tex`, lines 650–880, at the snapshot in [5]. The all-resonance subtraction and collapse are attributed to this source.